



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Cryptographic Failures & Data Protection

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Role: Cybersecurity Intern

Duration: 3 months

Assignment Submission Date: Monday, 6th October 2026

Day:26 (Twenty-Six)

Table of Content

1.A04:2025 CRYPTOGRAPHIC FAILURES.....	3
1.1 PART A — FUNDAMENTALS	3
1.2 PART B — PASSWORD & KEY SECURITY	5
1.3 PART C — CRYPTOGRAPHY	6
1.4 PART D — PRACTICAL ANALYSIS	7

1.A04:2025 CRYPTOGRAPHIC FAILURES

1.1 PART A — FUNDAMENTALS

1.What is cryptography?

Cryptography is a method of protecting information using mathematical techniques. It helps keep data confidential, secure, and protected from unauthorized access.

2. What is A04:2025 Cryptographic Failures?

Cryptographic failures happen when sensitive information is not protected properly. This can include weak encryption, insecure passwords, exposed keys, or unsafe communication.

3. What is sensitive data? Give four examples.

Sensitive data is information that should be protected from unauthorized access. Examples include passwords, personal information, financial information, and API keys.

4. Explain data in transit.

Data transit means information that is moving between two systems. For example, data moving between a browser and a website or between a mobile app and an API.

5. Explain data at rest.

Data at rest means information that is stored somewhere and is not currently moving. Examples include data stored in a database, hard drive, backup, or cloud storage.

6. What is encryption?

Encryption changes readable data into an unreadable form called ciphertext using an algorithm and a key. The data can later be decrypted by an authorized user with the correct key.

7. What is hashing?

Hashing changes input data into a fixed-format hash value using a hash function. It is mainly one-way process and is commonly used for integrity checks and password storage.

8. Explain the difference between encryption and hashing.

Encryption is designed so that data can be decrypted back into its original form using a key. Hashing is designed as a one-way process and is not normally reversed to recover the original data.

9. Why is Base64 not encryption?

Base64 is only an encoding method that changes the format of data. It does not protect or encrypt the information, so anyone can decode it easily.

10. Why should passwords never be stored in plaintext?

Plaintext passwords can be directly seen if the database is exposed. Passwords should instead be stored using secure password-hashing methods such as Argon2id, bcrypt, or PBKDF2.

1.2 PART B — PASSWORD & KEY SECURITY

11. What is password hashing?

Password hashing converts a password into a hash value before it is stored. This helps protect the original password if the password database is exposed.

12. What is a salt?

A salt is a unique random value added to a password before hashing. It helps make password hashes more difficult to attack using precomputed lists.

13. Why are unique salts useful?

Unique salts help ensure that the same password does not automatically create the same stored hash for different users. This provides better protection for password storage.

14. What is a cryptographic key?

A cryptographic key is a piece of security information used by cryptographic operations. It can be used for encryption, decryption, or digital signing.

15. What is key management?

Key management means securely handling cryptographic keys throughout their lifecycle. It includes generating, storing, protecting, rotating, and revoking keys.

16. Why is hardcoding a production encryption key in source code risky?

It can expose the key to developers, repositories, or anyone who gets access to the source code. It also makes changing or rotating the key more difficult.

1.3 PART C — CRYPTOGRAPHY

17. Explain symmetric encryption.

Symmetric encryption uses the same secret key for encryption and decryption. It is generally efficient and is commonly used for protecting data.

18. Explain asymmetric cryptography.

Asymmetric cryptography uses two related keys: a public key and a private key. It is commonly used for secure communication, authentication, and digital signatures.

19. What is TLS?

TLS stands for Transport Layer Security. It protects communication over a network by providing encryption, integrity, and server authentication.

20. Why is HTTPS important when transmitting sensitive information?

HTTPS uses TLS to protect information while it travels between the client and server. It helps prevent sensitive data such as passwords and personal information from being exposed during transmission.

1.4 PART D — PRACTICAL ANALYSIS

Analyze this fictional application: Login Transport: HTTPS Passwords: Stored as Plaintext
Encryption Key: Stored in Public Source Code Session Token: 10001, 10002, 10003...
Certificate Validation: Enabled

Answer:

1. Identify the positive security controls.

The application uses HTTPS to protect data in transits and has certificate validation enabled. These controls help provides secure communication between the client and server.

2. Identify every cryptographic problem.

The main problems are:

Passwords are stored in plaintext.

The encryption key is exposed in public source code.

Session tokens are sequential and predictable.

3. Which OWASP Top 10:2025 category applies?

The main category is A04:2025 – Cryptographic Failures because sensitive information and cryptographic keys are not being protected properly.

4. Why is plaintext password storage dangerous?

If the database is exposed, attackers could directly see the user's actual passwords. This could also put users at risk if they use the same passwords on other accounts.

5. Why is the hardcoded key a problem?

A key store in public source code can be exposed to unauthorized people. Once exposed, the key could potentially be misused to access or protect sensitive information.

6. Why are sequential security tokens weak?

Sequential tokens such as 10001. 10002. 10003 are easy to predict. Security tokens should use strong, unpredictable random values so they are harder to guess.

7. Give three defensive recommendations.

I. Store passwords using a secure password-hashing method with unique salts.

II. Keep encryption keys in secure key or secret management system instead of public source code.

III. Use cryptographically secure random values for session tokens.